

A Case Study: Data Science Salaries 2023 Visualization

Dataset Description

The dataset I picked is: Data Science Job Salaries

- Link: (<https://www.kaggle.com/datasets/arnabchaki/data-science-salaries-2023>)

I chose this dataset as it shows job market trends, in hopes of giving some insights to my future careers.

Initial Questions

1. Who gets the best pay? What position or job title should I apply to?
2. What is the correlation between the mode of employment (e.g., part-time/full-time, remote/on-site) and salary? Does company encourage workers to spend time in office rather than at home?
3. Do larger companies tend to employ individuals with more experience?
4. How did the COVID-19 pandemic (2020–2022) affect workers? Did it make it harder to get well paying jobs?

Getting Data into Observable

`ds_salaries.csv`



To address the broad scope of Question 1, analysis of the question will be divided into two distinct visualisations to provide better charts:

Q1a) Experience Level and Salary Correlation: A bar chart will be used to illustrate the relationship between experience levels and their salary range.

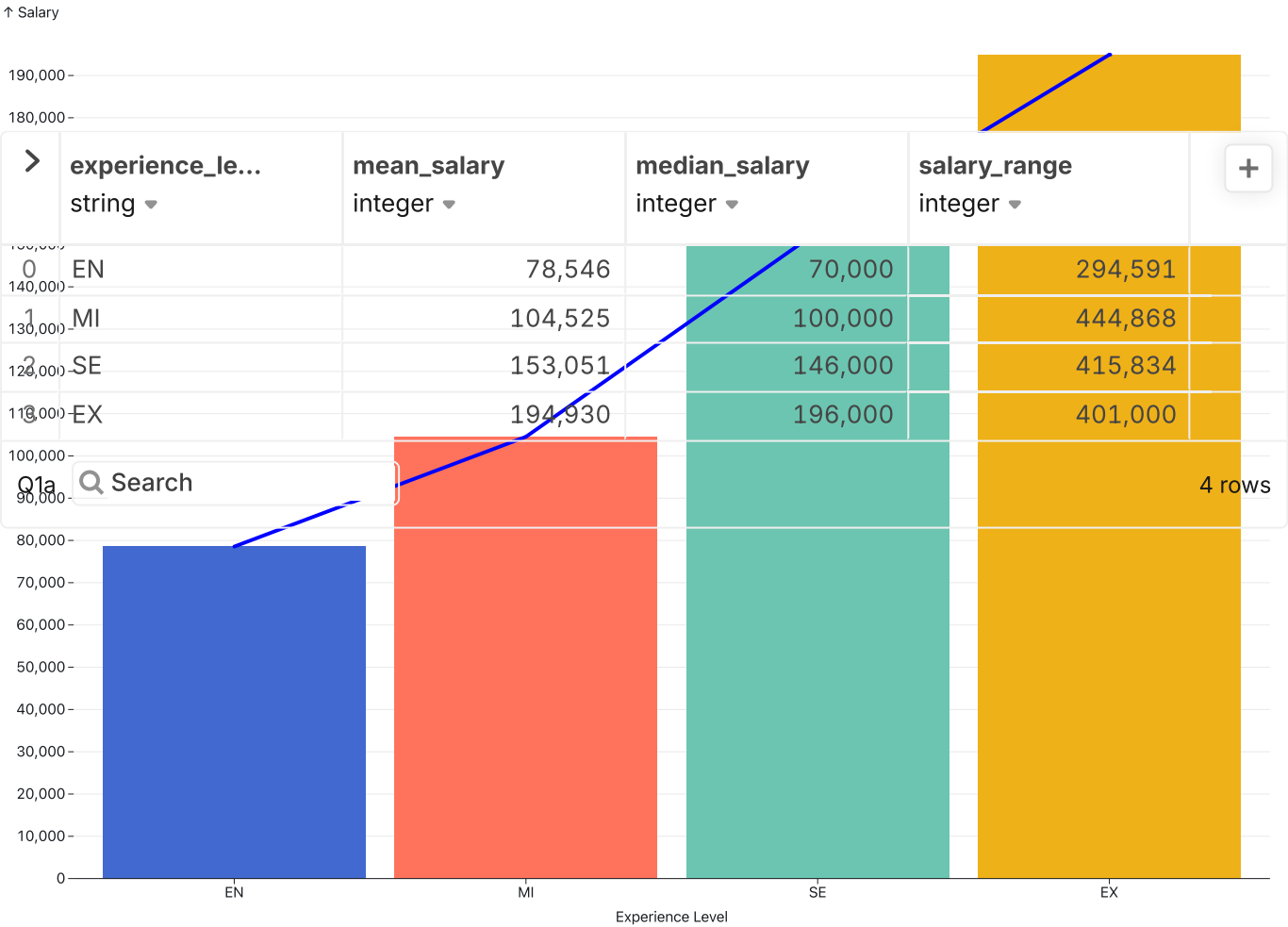
Q1b) Job Title and Salary Correlation: A separate bar chart will draw the relationship between top 5 best paying job titles and their salary levels, highlighting roles that are associated with higher income.

This approach ensures that both the impact of experience level and the influence of job titles are considered and are effectively visualized.

Bar Chart to visualise Salary distribution grouped by experience level

The 4 levels ranked from least experience to most being:

- EN: Entry Level
- MI: Mid-Level
- SE: Senior Level
- EX: Executive Level



The dataset was pre-processed to focus on key variables relevant to this question. Specifically, the following columns were retained: experience_level, mean_salary, median_salary, and salary_range. Experience levels were sorted in ascending order (Entry Level (EN), Mid-Level (MI), Senior-Level (SE), and Executive-Level (EX)) to ensure a chronological progression for the viewer. This process guides a structured evaluation of

<https://observablehq.com/d/be3208ebcffa1482>

chronological progression for the viewer. This process guides a structured evaluation of salary patterns across different experience tiers.

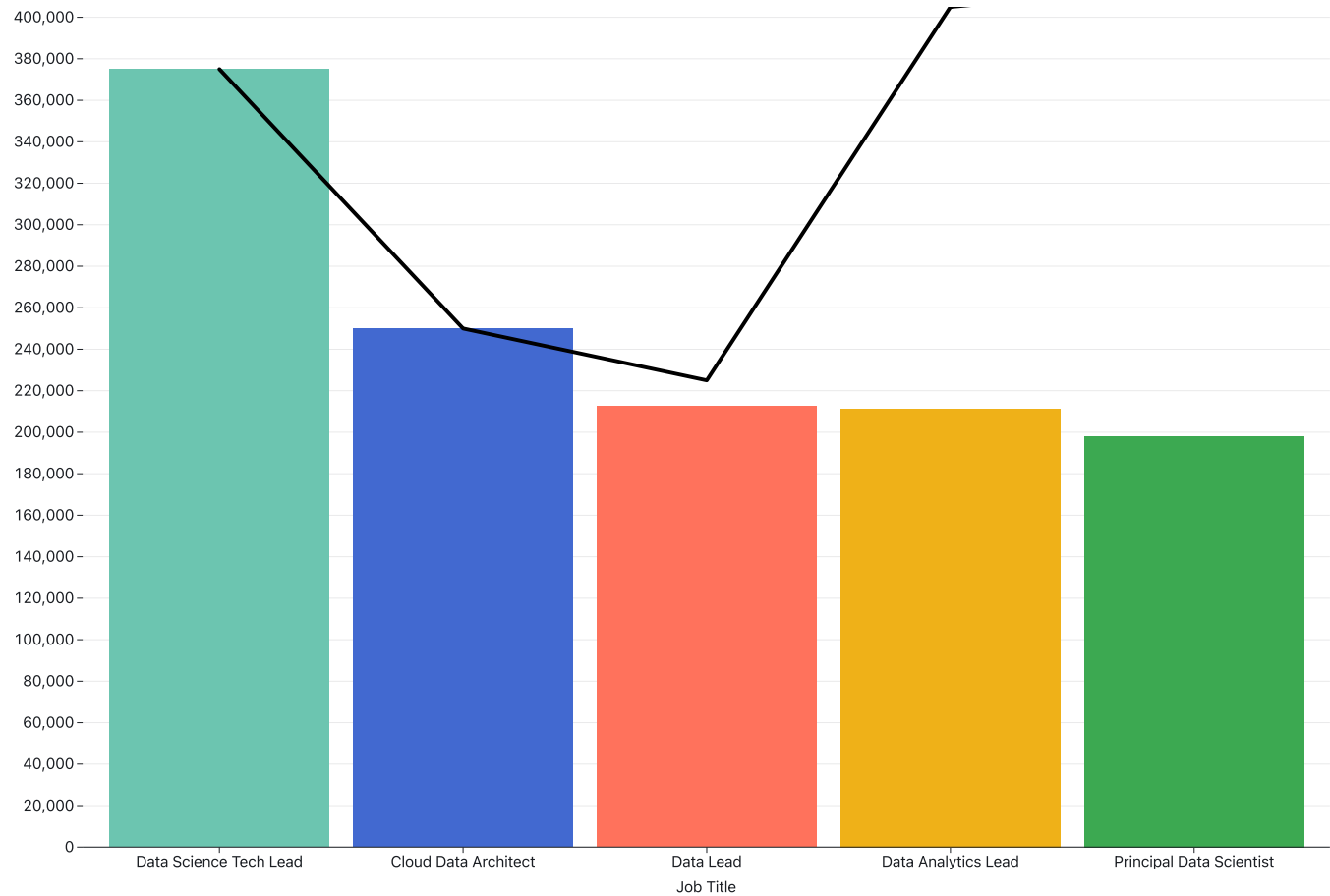
To represent the relationship between experience level and salary, a combination bar chart and line plot was selected. The bar chart was used to show the mean salary for each experience level, while the line plot placed on top displayed the growth in mean salaries across these levels. This duo allows me to convey at the same time interger values (salary) and the overall trend, the more experience the more the salary. Key Features

1. Bar Chart for Categorical Comparison: o The bar chart provided a clear and intuitive means of comparing mean salaries across categorical experience levels. o Distinct colour coding was applied to bars corresponding to each experience level, improving visual differentiation and enabling quick identification of categories.
2. Line Plot for Trend Analysis: o A trend line was overlaid on the bar chart to highlight the progression of mean salaries. o The line was styled in bold to emphasize the upward trend, making it easier for viewers to identify the correlation between experience and salary. Benefits this Approach:
3. The bar chart delivers a categorical snapshot of mean salaries, which is essential for immediate comparisons.
4. The trend line enhances this by illustrating the broader progression of salaries, aiding in the identification of patterns.
5. And together, these elements create a visualization that balances detail with clarity, enabling effective communication of both distribution and trends. Conclusion This analytical approach and visualization methodology provided an insightful representation of salary progression by experience level. Sorting experience levels logically and leveraging a hybrid visualization ensured that both individual salary values and overarching trends were effectively communicated. Colour coding and line emphasis further improved interpretability, making the analysis visually engaging and easy to understand.

Q1b)

Select the top 5 most well paid job

↑ Mean Salary

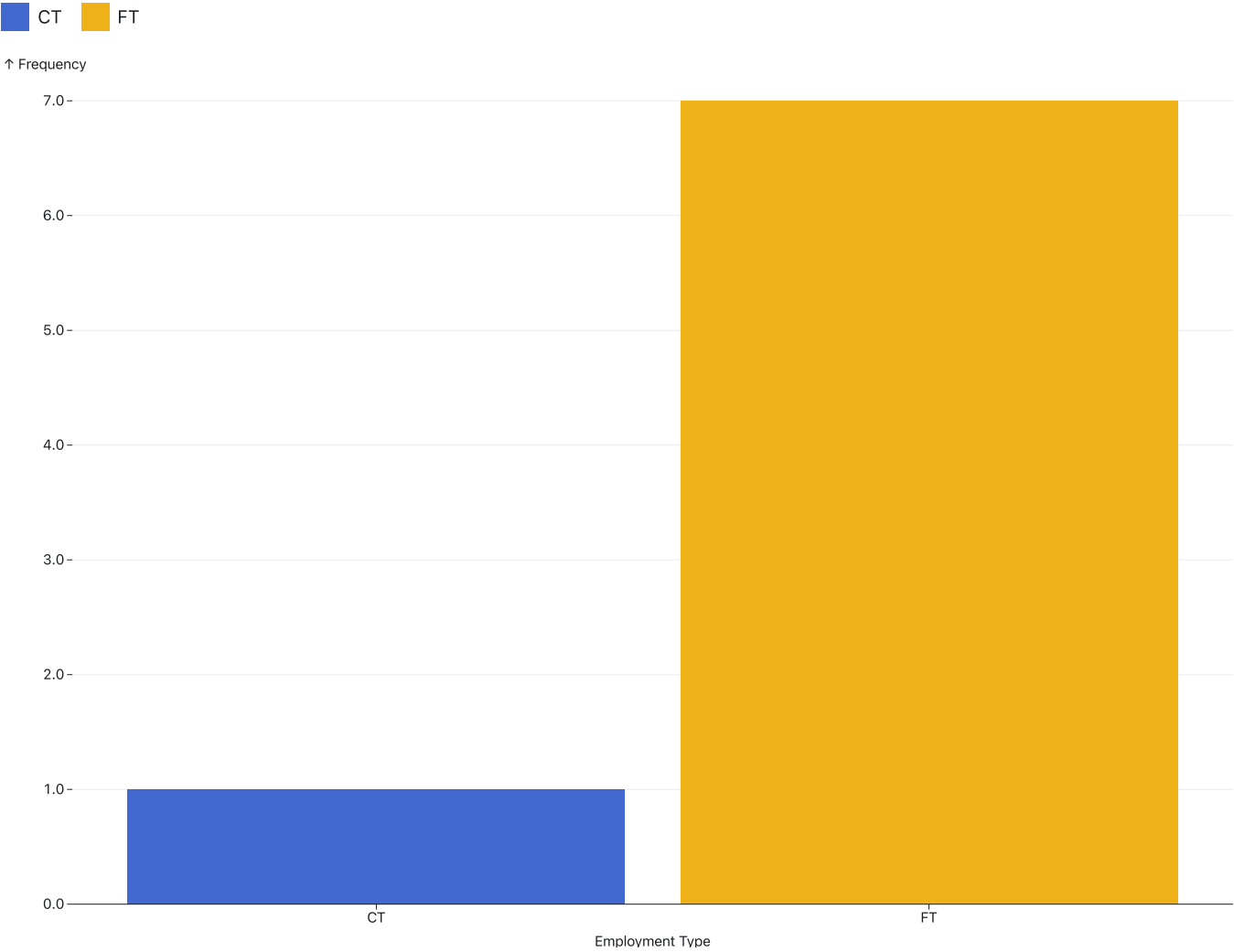


>	job_title	mean_salary	max_salary	
	string ▾	number ▾	integer ▾	+
0	Data Science Tech Lead	\$375,000	\$375,000	
1	Cloud Data Architect	\$250,000	\$250,000	

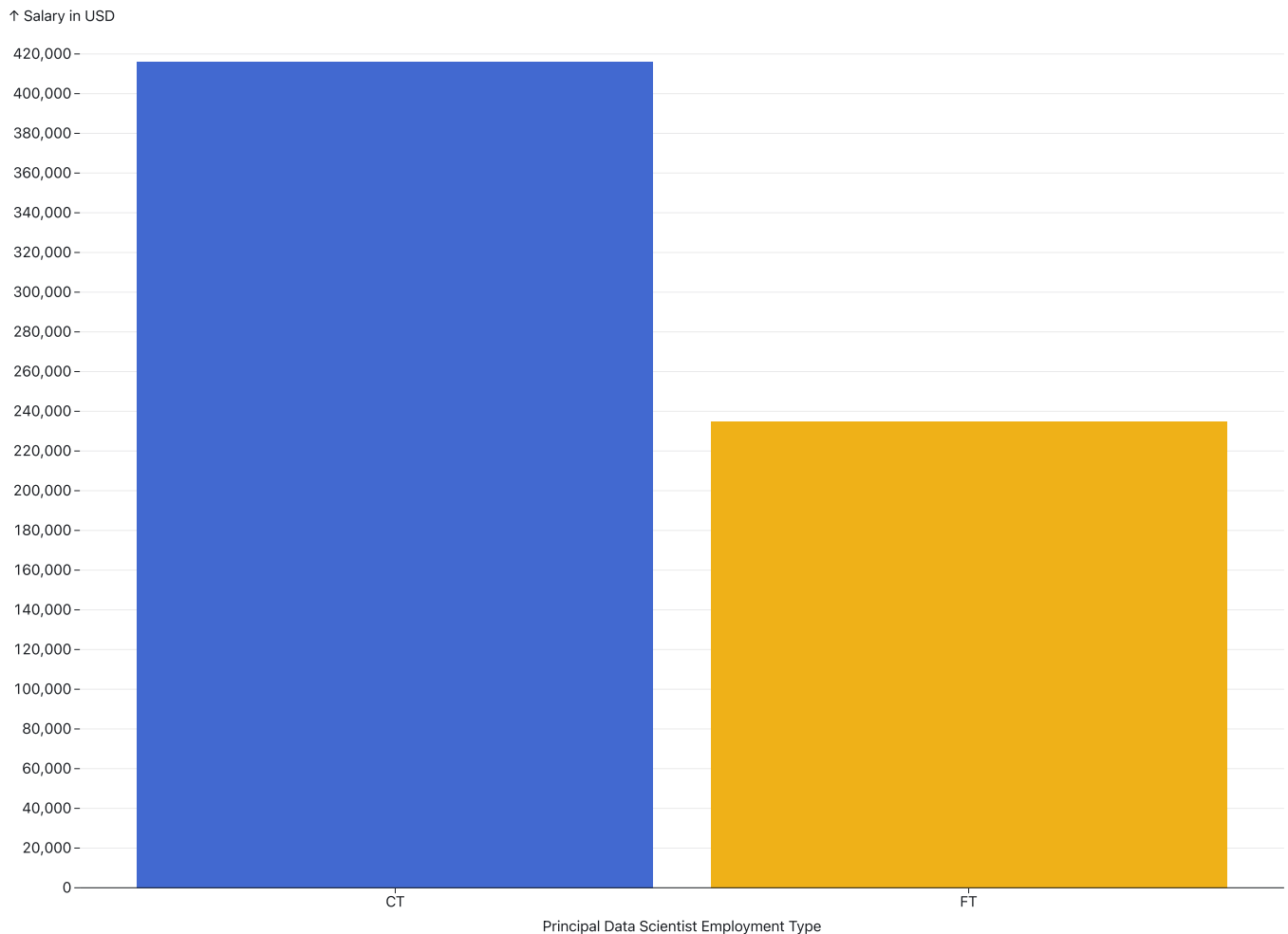
Q1b) Extension: Investigation of Principal Data Scientist pay

Why is there such a pay difference between the average and max pay of Principal Data

	string ▾	number ▾	integer ▾	
2	Data Lead	212,500	225,000	
3	Data Analytics Lead	211,254.5	405,000	
4	Principal Data Scientist	198,171.125	416,000	
Q1b	Search			5 rows



CT FT



Question 1b aims to analyse salary trends across top job titles offers critical insights into compensation structures and variability, using mean and maximum salary as focus stats. As expected R has been used for data selecting the values from the greater dataset. Visualisations were created in Observable, with a combined bar chart and line plot providing an easy comparison between salary distributions of the different work title. The bar chart highlights mean salaries across job categories, doing a good job of showing the best paying titles at a glance. At the same time, the line plot reveals maximum salary values, drawing out more stuff that may not be apparent in the bar chart. This two-layered visualisation effectively balances simplicity and depth and is suitable for both rapid assessments and detailed analysis. The analysis revealed significant differences between average and maximum salaries for specific roles, such as “Principal Data Scientist”. These discrepancies suggest that pay for these positions is likely to be influenced by external factors such as year of employment, geographic location, or experience level. For

```
Plot.plot({  
  width: 1400,  
  height: 1000,  
  marginLeft: 80,  
  marginTop: 50,  
  marginBottom: 50,  
  x: {  
    label: "Job Title",  
    domain: Q1b.map(d => d.job_title),  
  },  
  y: {
```

```
    grid: true,
    label: "Mean Salary"
  },
  marks: [
    Plot.barY(Q1b, { x: "job_title", y: "mean_salary",
                     fill: "job_title", tip: true }),
    Plot.line(Q1b, {x: "job_title", y: "max_salary",
                    stroke: "black", strokeWidth: 4}),
    Plot.ruleY([0])
  ],
  style: {
    fontSize: "16px"
  }
})
```

with annotations, geographic filtering, or additional contextual data could yield even richer insights.

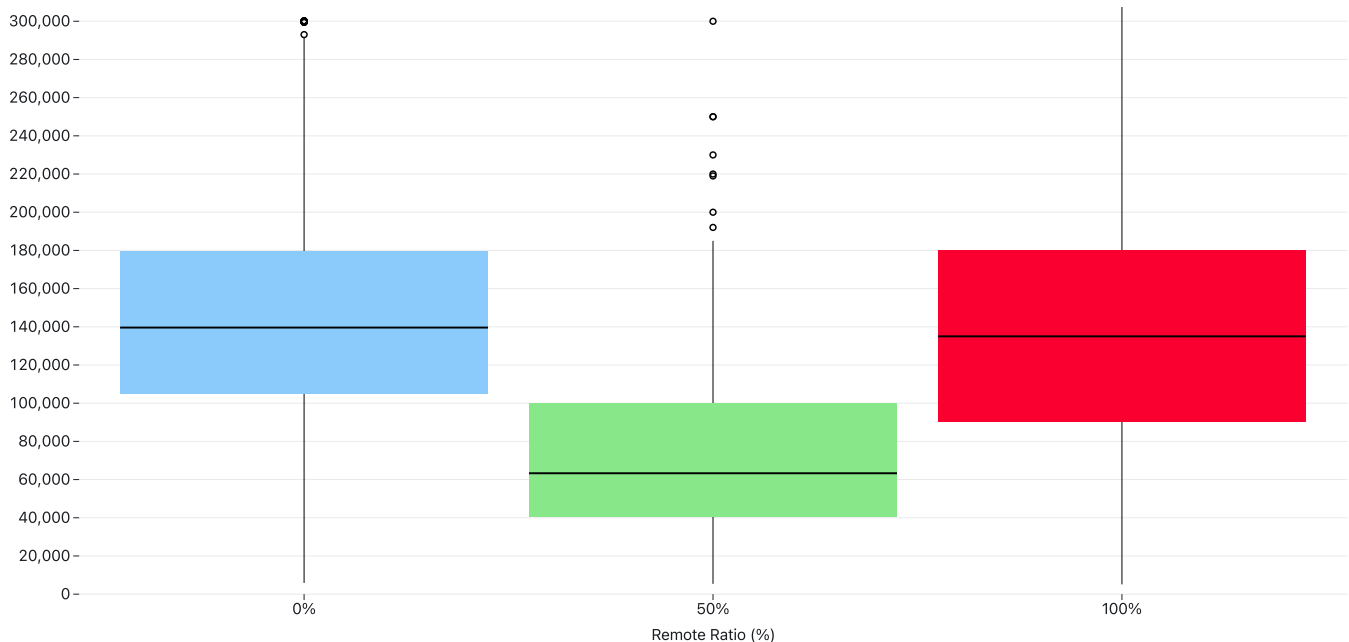
Question 2

✓	work_year integer ▾ 	experience_le... string ▾ 3 categories	employment_t... string ▾ 2 categories	salary_in_usd integer ▾ 	er + y string ▾ 3 categ
0	2,020	SE	FT	148,261	DE
1	2,021	SE	FT	220,000	US
2	2,021	EX	CT	416,000	US
3	2,021	SE	FT	173,762	DE
4	2,021	SE	FT	235,000	US
5	2,021	MI	FT	151,000	US
6	2,022	SE	FT	155,499	DE
7	2,023	SE	FT	85,847	ES

Q1be

8 rows





The visualisation for Question 2 is conducted using a box plot to compare the salaries of offsite, onsite, and hybrid employees across the three remote working ratio categories: 0% (onsite), 50% (hybrid), and 100% (offsite). While the limited number of ratio ranges (three) makes it challenging to identify a clear correlation between remote work ratios and salary trends, some observations can still be drawn.

The box plot reveals that both fully onsite (0%) and fully offsite (100%) workers tend to have similar median salaries, hovers around \$140,000. This similarity suggests that the salary impact may not be directly tied to the percentage of remote work for these two extremes. However, the hybrid (50%) category notably displays the lowest median and average salary compared to the other two groups. This disparity highlights that hybrid roles may not be as well-compensated, potentially due to differences in job roles or responsibilities associated with this working arrangement.

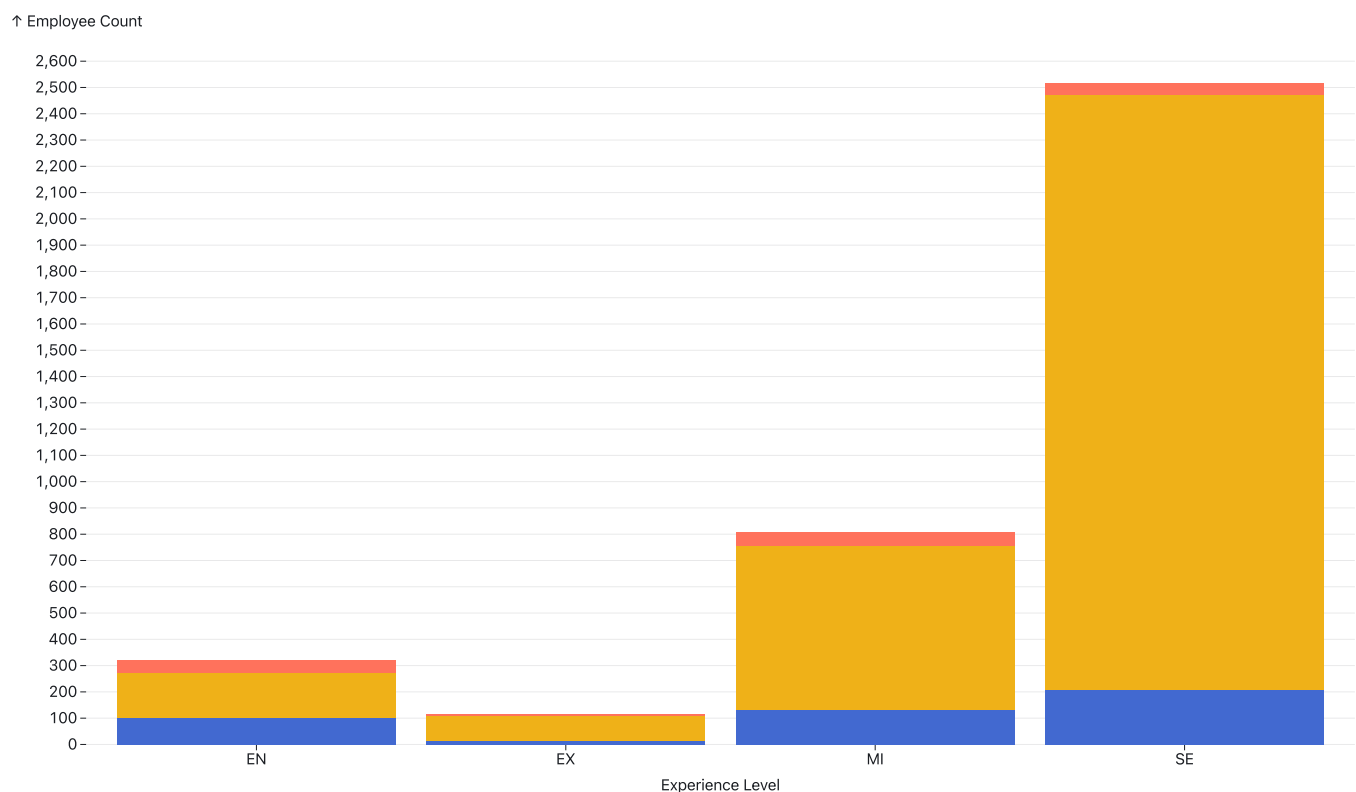
In terms of salary variability, fully offsite (100%) roles exhibit the largest range, with some salaries extending well beyond \$400,000. This range indicates that certain offsite positions are highly lucrative, likely due to senior-level responsibilities or specialised roles that allow remote work flexibility. Onsite (0%) roles show a more moderate salary range, with fewer high outliers, suggesting standardised pay structures. The hybrid (50%) group has the narrowest range, indicating minimal variability in pay and potentially reflecting a more uniform pay structure for these roles.

Question 3

Plot size of the company and Plot experience of employees? Investigate whether larger companies tend to employ individuals with more experience.

Company Size

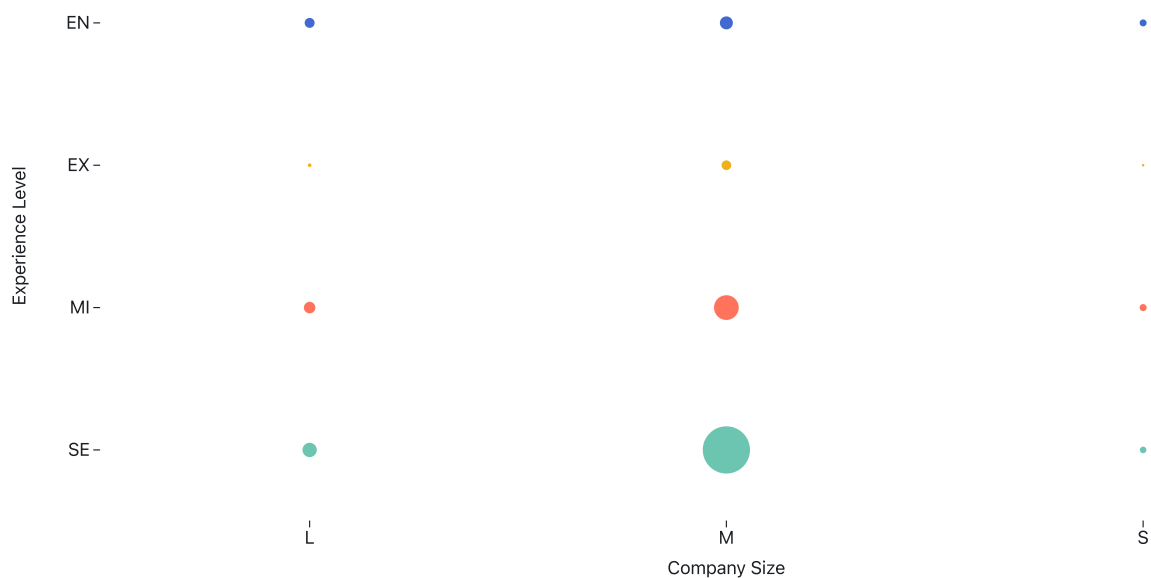
- S: Small
- M: Medium
- L: Large



A significant proportion of the dataset's employees belong to the Senior-level (SE) category, which accounts for over 2,500 Data Scientist, making it by far the largest category. This distribution highlights the dominance of senior-level professionals within the data, likely reflecting their critical role in driving organizational strategy and execution. Mid-level (MI) employees represent the second-largest group, with a population of around 800. which is surprising when compared to the 2,500 employees in the Senior-level. This disparity might point to either a bottleneck in career progression or the strategic hiring of senior professionals over mid-level employees. Entry-level (EN)

accounts for around 800 members which is a reasonable intake of fresh talent, which could be indicative of companies investing in long-term employee development. On the other hand, the Executive-level (EX) group, as expected, is the smallest, reflecting the typical hierarchical structure where leadership roles are fewer in number.

Additionally, it is evident that mid-sized companies employ the most people across all experience levels, which may indicate their capacity for balanced recruitment across seniority levels. The overall distribution raises intriguing questions about the career pathways and structural needs within organisations.



For Question 3, a bubble chart was selected to visualise the relationship between the number of employees across company sizes and their experience levels. This type of chart can represent three distinct dimensions, making it a powerful tool for analysis. The bubble size reflects the total number of employees in each category, while different colours indicate experience levels (Entry-level, Mid-level, Senior-level, and Executive-level). The x-axis represents company size, categorised as small, medium, and large, adding another dimension to the visualisation. Such a multidimensional analysis is not achievable with traditional bar or line charts. While this approach provided an interesting visual, it fell short in accurately capturing the true situation due to biases within the dataset. Specifically, the mid-sized company data displayed a highly imbalanced ratio, skewing the results. Furthermore, the Executive-level category had too few employees at only 6 in small companies and when compared to 2266 senior engineer in medium size companies. This meant that the corresponding bubbles were nearly invisible on the chart, making it difficult to extract meaningful insights. These issues severely limited the effectiveness of the bubble chart in conveying the nuances of the dataset. Given these challenges, a different approach was needed to ensure a more balanced and interpretable visualisation. Instead of focusing on absolute numbers, shifting to a ratio-based analysis for each company size might offer better insights. This change would allow for clearer comparisons across experience levels, regardless of the imbalance in raw employee

counts. Future visualisations could incorporate normalised metrics, such as the percentage of employees within each experience level per company size, to mitigate the

>

	salary_in_usd integer ▾	remote_ratio integer ▾	
0	85,847	100	
1	30,000	100	
2	25,500	100	
3	175,000	100	
4	120,000	100	
5	222,200	0	
6	136,000	0	
7	219,000	0	
8	141,000	0	
9	147,100	0	
10	90,700	0	
11	130,000	100	

Q2

🔍

Search

3,755 rows

EN

EX

MI

SE

↑ Ratio of different experience workers



Building on the insights from the previous charts, I recognised the need for a more refined dataset to uncover deeper trends. To achieve this, I used the formula

- $\text{Ratio} = \text{Number of Employees at X Level} / \text{Total Employees in the Company Size}$

to calculate the proportion of employees at each experience level within each company size. This updated dataset allow for an improved stacked bar chart, which provides a clearer visualisation of the distribution of workforce experience levels across small, medium, and large companies.

The chart reveals surprising trends in hiring patterns. Contrary to initial assumptions, large companies do not exhibit a disproportionately high ratio of Executive-level (EX) employees. Instead, their hiring practices closely align with those of medium-sized companies, with both focusing primarily on Senior-level (SE) employees. Senior professionals dominate these organisations, suggesting that both large and medium-sized companies prioritise experienced talent for strategic roles and leadership responsibilities.

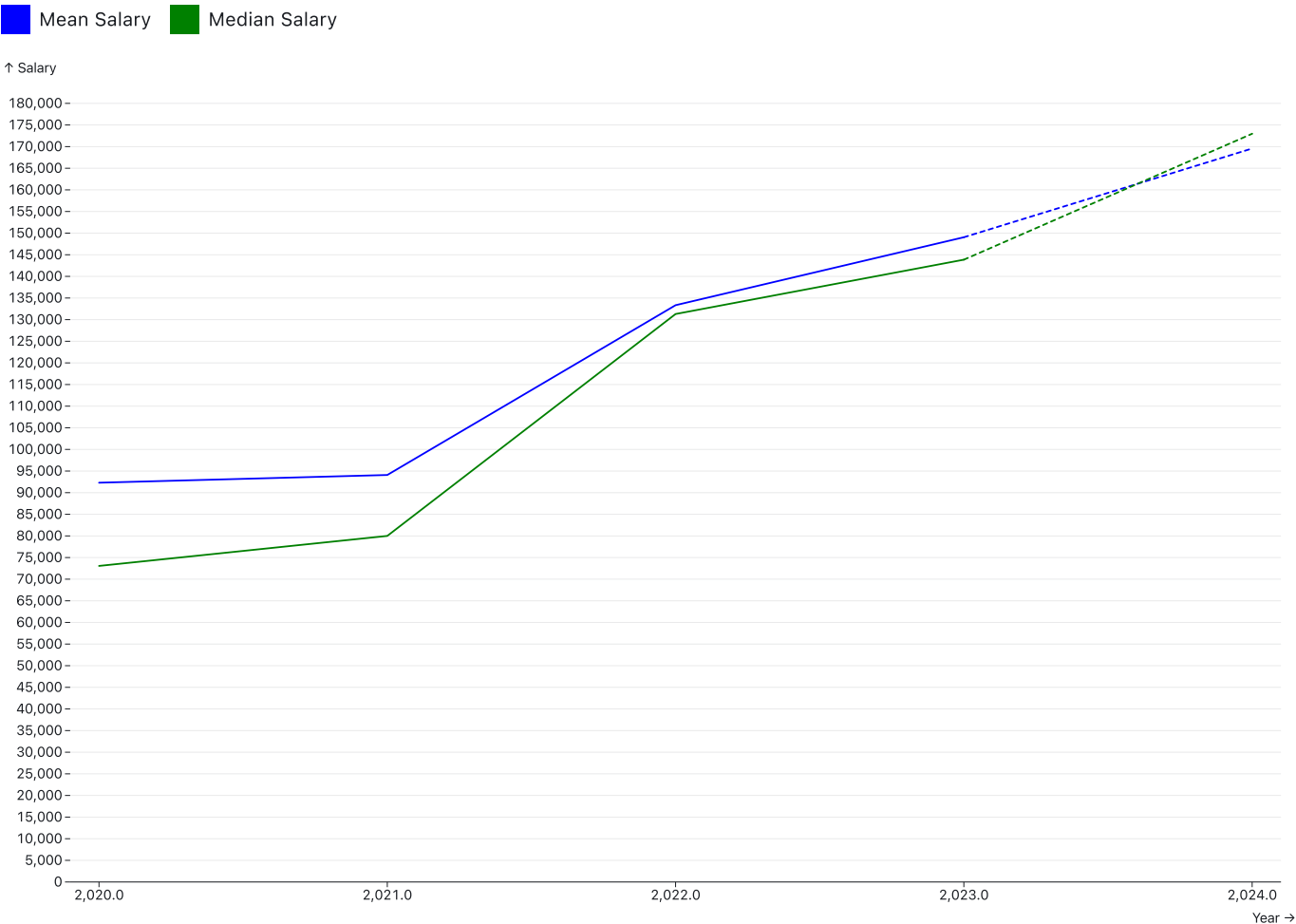
In contrast, small companies demonstrate unique hiring trends. They employ the highest proportion of Entry-level (EN) employees, with 33% of their workforce consisting of less experienced professionals. This is significantly higher than the 5% observed in medium-sized companies and the 22% in large companies. This trend suggests that smaller organisations may prioritise the recruitment of fresh talent, likely due to cost constraints or a focus on growth and development. These companies may also provide a starting point for less experienced employees to gain skills and progress in their careers.

Mid-level (MI) employees also represent an interesting aspect of the workforce distribution. Small companies employ 34% of their workforce at the mid-level, which is slightly higher than the 30% observed in large companies and significantly more than the 20% in medium-sized organisations. This suggests that small companies maintain a more balanced workforce structure, likely to address a variety of operational demands without leaning heavily on senior professionals. This balance may reflect the more dynamic and diverse role requirements typical of smaller firms.

Overall, this distribution highlights how the experience levels of employees vary significantly by company size. Large and medium-sized companies tend to emphasise senior professionals, leveraging their expertise for strategic objectives. Small companies, on the other hand, focus more on entry- and mid-level employees, likely reflecting a cost-effective, growth-oriented approach. These differences underscore the varying priorities and operational needs of organisations of different sizes, with larger firms focusing on stability and leadership while smaller firms aim for flexibility and development.

Question 4

4a) Worker's pay



For Question 4a, I selected a two-line graph to visualise the trends of mean and median

▼	company_size string ▼	experience_le... string ▼	employee_count integer ▼	+
	3 categories	4 categories		
0	were then incorporated into the visualisation, with the forecasted segment from 2023 to			
1	2024 represented as a dotted line. This difference should help viewers separate the			
2	predicted values from the actual data, ensuring clarity while highlighting the potential			
	salary trends for the coming year.			

▼

	string ▼	string ▼	integer ▼	
	3 categories	4 categories		
4	M	EN	171	
5	M	EX	95	

Q3

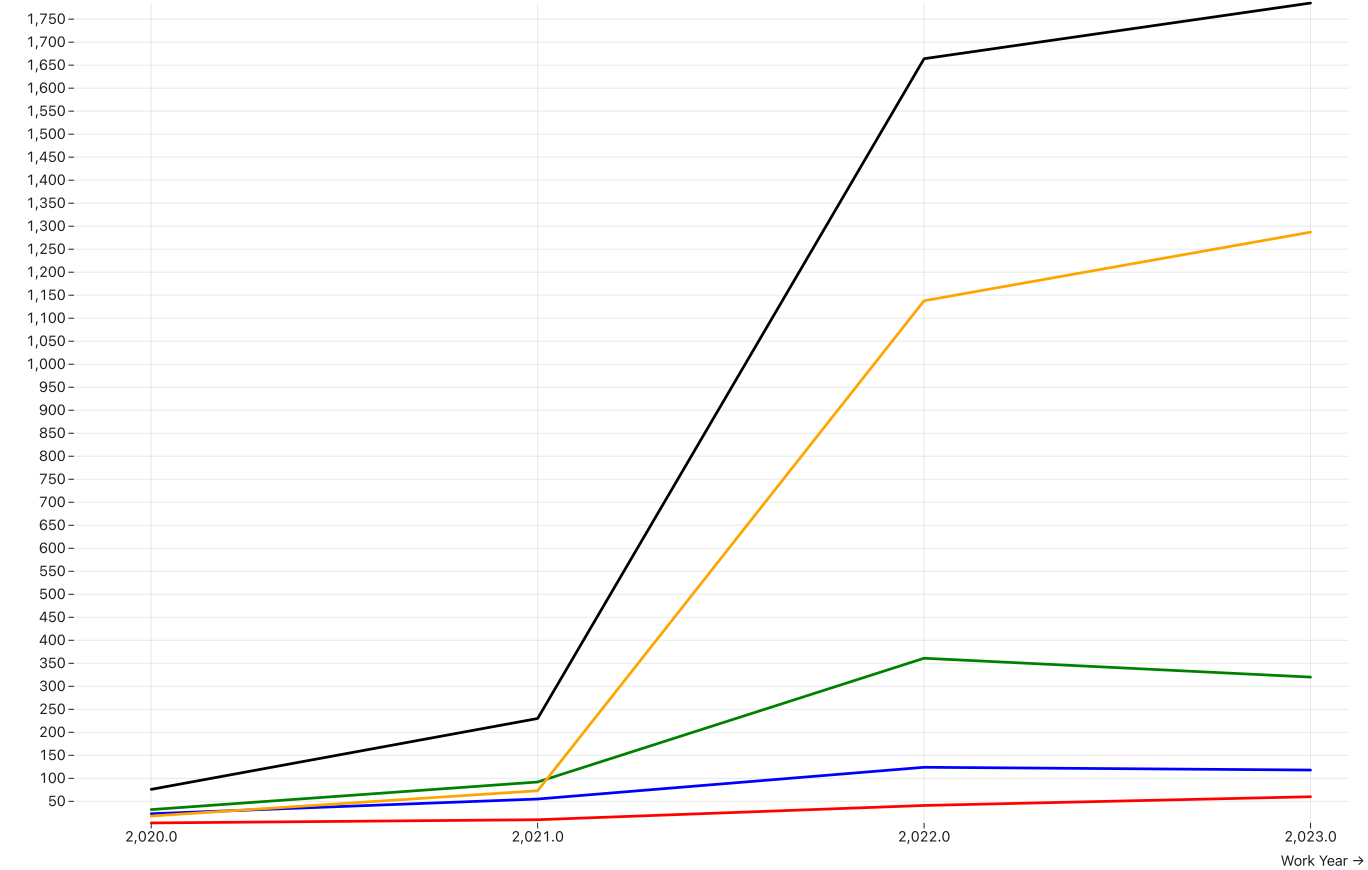
🔍

Search

12 rows

EN MI SE EX Total

↑ Worker Count



In Question 4b, I utilised a multi-line chart to visualise the distribution of data science work opportunities across four the experience levels (refer to question 1a), alongside the total number of roles available for each year (depicted by the black line). Each experience level is represented by a distinct color: blue for EN, green for MI, orange for SE, and red for EX, making it easy to differentiate the trends. This design effectively captures the growth trajectory of data science opportunities from 2020 to 2023.

A key observation from the chart is the significant rise in total data science work opportunities over the years. The total roles, illustrated by the black line, surged from 230 in 2021 to 1,664 in 2022, reflecting a remarkable increase of over sevenfold. By 2023, this number had further climbed to 1,785. This exponential growth highlights the soaring demand for data science professionals across all experience levels, likely driven by the accelerated digital transformation following the COVID-19 pandemic.

Breaking down the growth by experience levels, it is evident that Senior-level (SE) roles have consistently dominated the workforce, as indicated by the orange line. These roles show steady growth over the years, reflecting organisations' emphasis on hiring experienced professionals to lead projects and mentor teams. Mid-level (MI) positions, depicted in green, also exhibit a noticeable increase, peaking in 2022 before slightly declining in 2023, possibly due to companies' focus on either entry-level hires or senior professionals. Entry-level (EN) roles, shown in blue, have a steady but modest growth pattern, indicating a slower intake of fresh talent compared to experienced hires. Executive-level (EX) positions, represented by the red line, remain the least prevalent, reflecting the naturally limited number of high-level strategic roles in most organisations.

Benefits:

1. Excellent for showing trends over time, making it easy to observe the overall growth in opportunities and the relative growth patterns for different experience levels.
2. Multiple lines enables a direct comparison between the different experience levels and the total number of roles.
3. Effectively display the various data without making it too overwhelming to the viewer.

Overall, the line chart effectively highlights the rapid growth of data science opportunities while providing insights into how these roles are distributed across experience levels. However, the inability to incorporate all job titles and predict 2024 values due to exponential growth remains a limitation of this approach.

Reflection

The development process for this coursework has been both iterative and intriguing, refining the dataset and changing the plots to suit my needs. Starting with a clear objective of understanding data science job market opportunities and trends across different experience levels, the process evolved with new challenges and discoveries that

shaped the final output. One of the most important aspects of this coursework process was data preparation and transformation. The raw dataset required extensive data pre-processing in R to ensure the data I need is there. Calculating key metrics such as mean_salary, median_salary, salary_range, and worker_count across years and experience levels involved various interesting techniques learnt during the labs and lectures. The inclusion of experience-level-specific data brought an additional layer of quality, allowing for better insights and more better comparisons.

The selection of visualisation charts is fundamental to gaining a clearer understanding of the dataset. Initially, relying solely on bar charts and line charts for the entire dataset proved to be overly simplistic. However, as I progressed through the coursework, I discovered more effective visualisation methods, such as stacked bar charts and bubble plots, which better captured the complexities of these data. These choices highlighted the iterative nature of this development process, where initial question is continuously refined to create new insights and question.

A significant challenge was balancing clarity and complexity. For instance, including all job titles or detailed breakdowns initially seemed appealing but quickly led to clusters and unreadable charts. Simplifying the visualisations to focus on total opportunities and key experience levels ensured the story remained clear and impactful. Similarly, while predicting the 2024 values for total roles and salaries was considered, the exponential growth in recent years introduced uncertainties that made such predictions speculative. A decision was made to refrain from forecasting for 2024, prioritising data integrity and reliability over potentially misleading projections.

The Interactivity provided using Observable, such as tooltips, allows user to hover on top of the each datapoints of the graph to get the exact number making it more engaging and user friendly. Features like various colour coding and legends further improved the

▼ company_size string ▼		experience_le...	employee_count	ratio	+
3 categories		4 categories	integer ▼	number ▼	
					
adaptable, combined with iterative refinements and thoughtful design choices, ensured		that the final visualisations effectively communicated the key insights.			
2	L	MI	134	0.295	
3	L	SE	207	0.456	
4	M	EN	171	0.054	
5	M	EX	95	0.03	
6	M	MI	621	0.197	
7	M	SE	2,266	0.719	
8	S	EN	49	0.331	
9	S	EX	6	0.041	
company_experience_ratio		Search			12 rows

▼	year integer ▼ 20202024	mean_salary number ▼ 80k160k	median_salary integer ▼ 60k160k	salary_range integer ▼ 410k450k	work_experience integer 0
0	2,020	92,302.632	73,065	444,293	
1	2,021	94,087.209	80,000	417,591	
2	2,022	133,338.621	131,300	425,835	
3	2,023	149,045.541	143,860	416,834	
Q4a	🔍 Search				4 rows

▼	experience_le... string ▼	year integer ▼	mean_salary number ▼	median_salary number ▼	sa + integer
	4 categories	 20202023	 50k210k	 40k210k	 200k
0	EN	2,020	57,511.609	45,896	
1	EX	2,020	139,944.333	79,833	
2	MI	2,020	87,564.719	78,395.5	
3	SE	2,020	137,240.5	118,552	
4	EN	2,021	54,905.255	55,000	
5	EX	2,021	186,128	151,833.5	
6	MI	2,021	82,116.935	72,606	
7	SE	2,021	126,085.356	120,000	
8	EN	2,022	77,006.024	61,252	
9	EX	2,022	188,260.293	187,200	
Q4b Search 16 rows					

